

SUPPLEMENTARY MATERIAL

VII. BASELINE EXPERIMENTS WITH FULL CATEGORIES

Figure 6 presents the output results of the Ground Truth and models across various scenarios. It is evident that the majority of models exhibit a limited understanding of mining roads. However, models founded on the Transformer structure, featuring a substantial number of parameters and self-attention mechanisms, demonstrate improved performance. This dataset can offer a theoretical foundation for researchers engaged in edge detection, small target extraction, and drivable area detection within unstructured roads.

Table VI shows the performance of all categories in different models. In the same category, the number in bold black font represents the best result, and underlined represents the second best result. The metrics are mIoU scores, with "MTruck", "Soil", "Testing", "BStone", "ICCone", "CPole", "TPole", "WCar", "OTruck", "WBTruck" representing Mining Truck, Soft Soil, Road Testing Equipment, Blocking Stone, Ice Cream Cone(or Traffic Cone), Communication Pole, Telegraph Pole, Water Car, Oil Truck, WBTruck Wide-body Truck.

TABLE VI

THIS TABLE SHOWS THE PERFORMANCE OF ALL CATEGORIES IN DIFFERENT MODELS. IN THE SAME CATEGORY, THE NUMBER IN BOLD BLACK REPRESENTS THE BEST RESULT, AND UNDERLINED REPRESENTS THE SECOND-BEST RESULT.

Category	Segformer	Mask2Former	Swin	FCN	Segnext	Segmenter	UPerNet	deeplabv3+	U-Net
mIoU	58.09	<u>56.71</u>	44.37	42.24	36.76	32.89	32.85	32.21	28.25
Sky	<u>99.80</u>	99.81	99.74	99.69	99.55	98.81	99.72	98.90	99.43
Road	88.86	<u>88.65</u>	86.37	86.23	86.55	78.23	86.03	83.88	83.99
Massif	<u>84.83</u>	86.46	82.41	79.78	81.22	74.63	80.51	79.19	74.42
Rut	<u>43.38</u>	46.38	24.55	33.31	30.96	23.12	32.72	30.60	22.9
Edge	<u>68.96</u>	70.84	63.82	62.03	65.40	56.00	64.86	63.46	54.31
Brushwood	<u>62.98</u>	63.58	54.50	29.48	56.10	49.60	12.04	50.96	35.22
MTruck	90.05	<u>89.22</u>	80.85	74.16	69.20	76.09	81.37	73.81	68.75
Soil	<u>29.51</u>	52.68	19.40	2.05	9.54	6.71	10.37	4.16	1.54
Gravel	18.32	15.79	25.77	<u>20.86</u>	16.70	9.87	12.26	16.22	11.52
Puddle	59.91	63.29	58.46	60.31	39.73	45.11	<u>61.90</u>	42.75	43.82
Shovel	81.66	<u>73.87</u>	59.61	49.84	51.01	44.55	34.18	34.12	43.7
Testing	<u>79.54</u>	80.97	64.43	56.95	35.11	40.31	61.13	13.31	44.03
Engineering	48.07	<u>38.40</u>	19.05	26.88	22.62	31.12	17.93	19.03	22.05
Pushdozer	<u>49.30</u>	50.13	23.82	41.95	35.01	38.84	37.28	21.79	20.24
Truck	39.72	<u>31.50</u>	24.17	20.65	16.46	21.30	10.57	12.65	9.94
Car	49.44	47.40	42.11	47.27	39.25	39.03	27.91	<u>48.06</u>	28.12
Mound	7.53	10.84	11.12	10.60	7.71	11.91	<u>11.19</u>	8.35	8.49
Vegetation	<u>32.52</u>	33.4	28.17	29.54	26.79	15.42	<u>27.35</u>	18.67	6.23
BStone	34.23	<u>33.86</u>	16.78	24.70	11.67	1.65	17.91	6.58	7.81
Excavator	73.05	34.76	<u>60.15</u>	52.72	13.52	38.26	23.21	21.24	17.09
Construction	<u>67.32</u>	74.42	58.59	53.19	35.29	56.04	65.23	42.26	32.78
ICCone	52.13	<u>51.89</u>	34.27	33.71	22.05	4.32	0.00	4.81	0.00
Fence	34.09	10.55	16.36	<u>32.15</u>	10.96	0.88	7.69	6.92	12.87
Sign	<u>71.33</u>	73.12	58.05	49.12	38.08	41.32	38.85	27.74	28.83
Cable	<u>32.81</u>	33.50	14.66	19.70	11.25	0.94	8.13	0.00	5.42
CPole	63.58	<u>50.06</u>	27.54	37.62	34.89	0.00	0.00	0.00	0.00
TPole	<u>70.58</u>	77.41	43.77	18.56	44.38	2.13	7.74	22.79	17.27
WCar	<u>88.95</u>	95.33	38.68	42.87	24.45	40.57	19.30	23.39	26.48
OTruck	95.57	<u>95.29</u>	72.65	68.57	50.93	0.06	0.00	54.93	0.00
Human	<u>11.27</u>	15.49	3.55	0.00	0.50	5.47	0.00	0.00	0.00
WBTruck	71.42	<u>69.07</u>	62.20	44.9	52.73	67.16	60.94	67.79	48.54

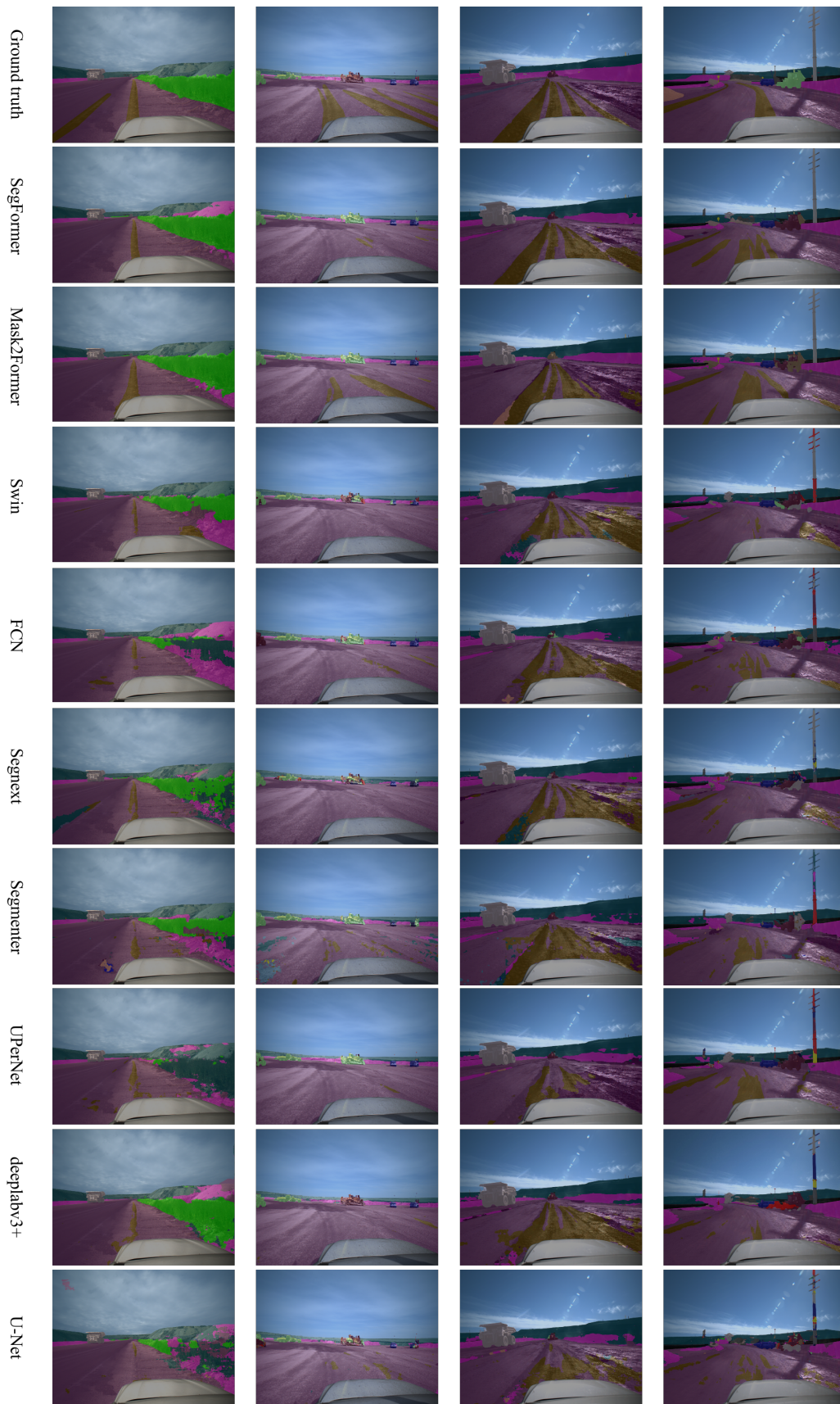


Fig. 6. Visual representation in several scenarios.